Chaoyi Huang

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Education

Shanghai Jiao Tong University Shanghai, China

B.S. in Computer Science, ACM Honors Class Expected June 2027

Selected coursework: Computer Vision (100/100), Introduction to Computer Systems (98/100), Principle and Practice of Computer Algorithms (97/100), Large Language Models (94/100).

Research interests

I study how robots learn to use unfamiliar tools and how a manipulation policy can use observations and memory to decide what to do next.

Research experience

The University of Hong Kong July 2026–present

Research Assistant, OpenDriveLab & JC STEM Lab Advisor: Prof. Hongyang Li

Proposed a method for tracking progress during multistage tool use. Compared ways to condition the policy on the current task stage and tested stage recognition on a physical robot. I also build real-robot setups and evaluate tool-use policies in simulation.

Shanghai Jiao Tong University Began June 2024

Research Assistant, Machine Vision and Intelligence Group, RHOS Lab
Advisors: Prof. Yong-Lu Li and Prof. Cewu Lu

Studied how to adapt a manipulation policy to an unfamiliar tool by identifying the part of the tool that performs the task.

Selected preprints

From Pixel to Poses: Object-centric Tool Manipulation Learning from Human Demonstrations
Bangjun Wang, Longyan Wu, Yukun Wei, Shenghe Shao, **Chaoyi Huang**, Wenze Cui, Zetong Xu, Hanlin Wu, Long Chen, Yi Ma, Hongyang Li.
arXiv:2609.35375 Project page

ProjDevBench: Benchmarking AI Coding Agents on End-to-End Project Development
Pengrui Lu, Shiqi Zhang, Yunzhong Hou, Lyumanshan Ye, **Chaoyi Huang**, Zixi Chen, Ji Zeng, Hantao Jiang, Pengfei Liu, Yiwei Wang, Ming-Hsuan Yang.
arXiv:2602.01655.

Selected projects

Contact-Aware 3D Object-Human Interaction PyTorch

Predicted human–object contact regions from images to resolve depth ambiguities in 3D reconstruction. Computer vision course project.

MxLang compiler Java

Built an ANTLR4-based compiler that translates a C++/Java-like language to RV32I assembly through an LLVM-inspired intermediate representation.

RISC-V kernel Rust

Implemented privileged-mode execution and process scheduling in a small operating system kernel.

RISC-V CPU Verilog

Implemented a Tomasulo-style processor with an instruction cache and branch prediction on an FPGA.

Ray tracer Rust

Built a path-tracing renderer with support for multithreading and OBJ models.

Teaching

Shanghai Jiao Tong University Teaching Assistant

Principle and Practice of Computer Algorithms (CS1952) Summer 2025
Great Ideas in Computer Science (CS1950) Fall 2024

Awards and honors

Zhiyuan Honors Scholarship, Shanghai Jiao Tong University 2023, 2024, 2025
Undergraduate Merit Scholarship (Class C), Shanghai Jiao Tong University 2024, 2025

Skills

Programming C++, Python, Java, Rust, Verilog
Tools PyTorch, ROS, SolidWorks, LaTeX
Languages Chinese (native), English (CET-6: 586/750)